

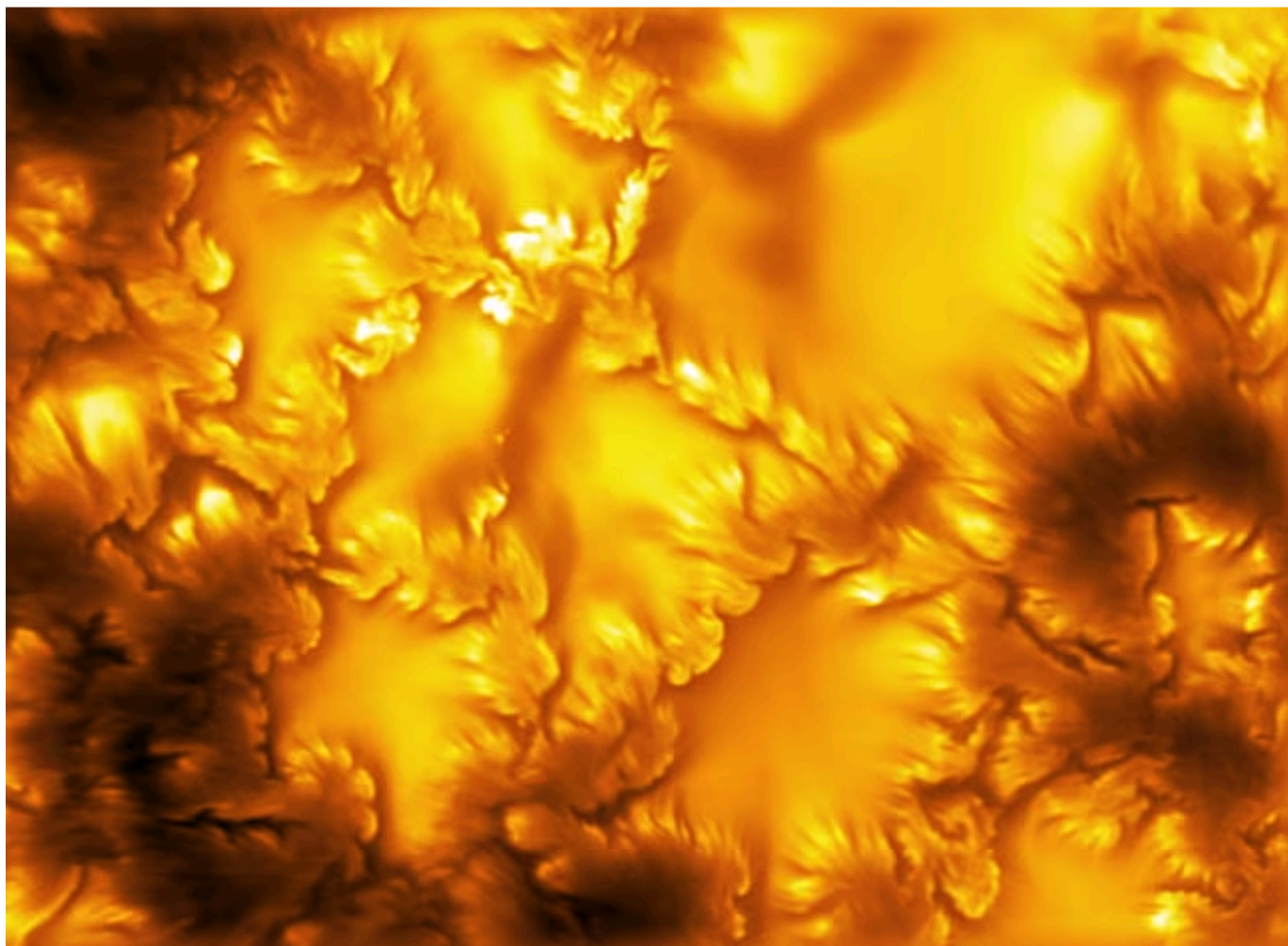
Eclipse 2026

НАУКА > КОСМОС • 5 МИНУТ ЧТЕНИЯ

Изображение Солнца в самом высоком разрешении, но с магнитными аномалиями

5 АВГУСТА 2026 ГОДА

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Изображение с самым высоким разрешением ...



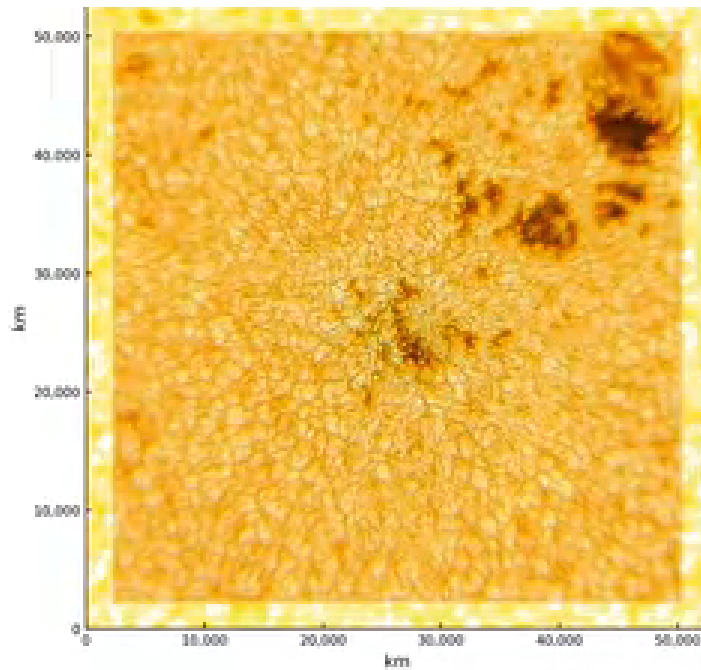
поверхности Солнца с высочайшим разрешением и обнаружил скрытый процесс, который влияет на солнечную активность.

Ученые использовали солнечный телескоп имени Дэниела К. Иноуэ Национального научного фонда, расположенный недалеко от вершины Халеакала, вулкана на гавайском острове Мауи, чтобы рассмотреть магнитно-активную область вблизи солнечного пятна. Солнечные пятна считаются очагами солнечной активности.

Снимки и таймлапс-видео позволяют по-новому взглянуть на сложную и динамичную фотосферу Солнца — видимую поверхность Солнца, которая представляет собой тонкий слой атмосферы, формируемый магнитными полями и потоками жидкой плазмы.

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Сочетание детальных снимков с компьютерным моделированием помогло исследователям совершить прорыв в области физики Солнца: они выявили характерные признаки небольших водоворотов на поверхности Солнца, которые могут напрямую влиять на жизнь на Земле.



Телескоп «Иноуэ» помог учёным рассмотреть мельчайшие детали движения на поверхности Солнца. *(Национальный научный фонд США / Национальное управление океанических и атмосферных исследований / AURA / MPS / HAO)*

Эти вихревые структуры, известные как неустойчивость Кельвина — Гельмгольца, могут объяснить некоторые загадки Солнца, например почему солнечная корона, или внешняя атмосфера, намного горячее поверхности.

Вихри также могут подпитывать накопление магнитной энергии Солнца, которая приводит к солнечным вспышкам и корональным выбросам массы. Когда эта солнечная активность направлена на Землю, она выбрасывает частицы, которые могут вывести из строя спутники, электросети и другую инфраструктуру связи.

помочь ученым понять поведение и активность Солнца, которые сложно предсказать.

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“Although theoretical models had suggested that the right conditions for Kelvin-Helmholtz Instability could exist in the photosphere, seeing these structures widespread across the surface was still a huge surprise,” lead study author Dr. David Kuridze, assistant astronomer at the National Solar Observatory in Boulder, Colorado, wrote in an email. “The vortex formation on the Sun has long been a central question in solar physics. For the first time, we have identified both their origin and their driving mechanism.”

A constantly shifting surface

Kelvin-Helmholtz instability occurs when two fluids traveling at different velocities move past one another, creating tiny perturbations that result in spiraling vortices, according to the study.

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formation and the atmospheres of large gaseous planets such as Jupiter and Saturn.

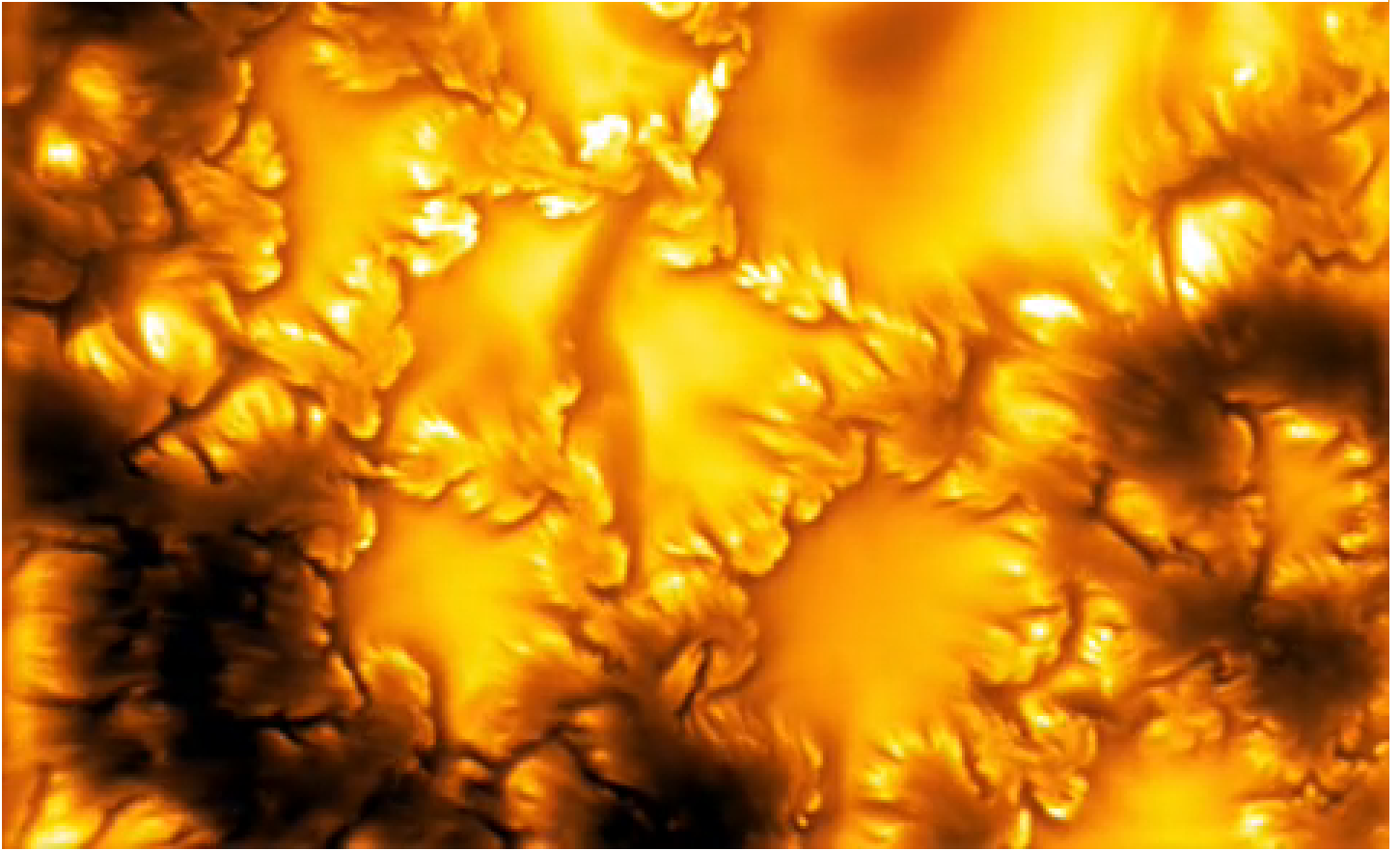
“A beautiful example of KHIs happens at the boundaries of Jupiter’s cloud bands, leading to vortices along the edges,” said Dr. Maria Weber, associate professor of physics and planetarium director at Delta State University in Mississippi. “The granddaddy of them all is the Great Red Spot.”

Weber was not involved in the new study.

The images from the Inouye telescope mark the first time the phenomenon has been seen on the sun.

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Scientists have long believed that the sun builds up stores of magnetic energy through “flux braiding,” when magnetic field lines twist together. The tension eventually becomes unstable, causing the tangle of magnetic fields to snap apart and then magnetically reconnect, releasing a burst of energy. But researchers never quite understood why these twisting patterns occur in the first place — but the new observations offer a possible explanation.



Small scale movements can be seen shifting on the sun's surface. *(NSF/NSO/AURA/MPS)*

Swirls occurring along the edges of magnetic regions across the solar surface could twist the fields together, the researchers realized. The formations also explain how heat makes its way to the sun's outer atmosphere.

“KHI is a really efficient way for the Sun to break big plasma flows down into smaller motions,” Kuridze said. “When you have KHI in the system, it makes it much easier to trigger an energy cascade toward tiny, microscopic scales and once energy reaches those micro-scales, it can easily be released as heat. Therefore, finding KHI across the solar surface gives us a very important missing piece of the puzzle.”

The magnetic vortices generated by Kelvin-Helmholtz instability effectively act like small-scale engines that can generate, transport and release energy throughout the solar surface, he said.

“We are still trying to piece together the full story about how the sun generates and sustains its magnetism, on all scales,” Weber said. “This work helps.”

The constantly moving whirlpools can act as energy stores for larger solar activity, such as flares and coronal mass ejections. But the sun is firing tiny nanoflares and other smaller-scale activity every second, Kuridze added.

“As it turns out, these micro-events are more influential for the Sun’s thermal and magnetic structure than rare, massive flares,” he said. “There is a strong consensus within the solar physics community that unlocking global solar behavior begins with understanding these micro-scales. Solar magnetism is fundamentally shaped by these small-scale events, which ultimately drive the space weather that impacts Earth’s technological infrastructure.”

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Dr. Nour Rawafi, project scientist for NASA’s Parker Solar Probe at the Johns Hopkins University’s Applied Physics Laboratory in Maryland, said the discovery is a breakthrough that could significantly improve how scientists understand of the dynamics of the sun’s lower atmosphere. Rawafi was not involved in the new study.

“It provides a possible mechanism for how the abundant mechanical energy generated in the lower solar atmosphere and convection zone is transformed and transferred into the Sun’s upper atmosphere,” Rawafi wrote in an email. “The widespread swirls and the associated Kelvin–Helmholtz instabilities make physical sense — we simply didn’t have the ability to observe the tiny solar structures where they occur so abundantly.”

The telescope observations very closely matched advanced computer simulations that were also run by the team during the study. The simulations enabled the team to interpret what they were seeing from the telescope’s data and tease out details that are more difficult to measure with observation alone.

Future observations could pinpoint the exact physical connections that enable predictions of large flares and solar storms.



The Daniel K. Inouye Solar Telescope is located in Maui. (*NSF/NSO/AURA*)

The observations were possible thanks to the Inouye telescope, which has provided awe-inspiring solar imagery since 2019.

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The telescope has a mirror with a 13-foot (4-meter) diameter, an adaptive optics system that corrects image blurring caused by Earth's atmosphere and instruments that can gather observations across the sun's photosphere, the second layer of its

Michael Wöger, senior scientist at the National Solar Observatory.

“The scale and performance of the Inouye combined with its unique instrumental capabilities make it a one-of-its-kind world class facility, enabling this kind of work,” Wöger said.

Rawafi believes this is just the beginning for what the telescope could reveal about the sun.

“It’s opening an entirely new window on the Sun, and I expect many more transformative discoveries to follow,” he wrote.



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